

Efficacy of metformin in patients with breast cancer receiving chemotherapy or endocrine therapy : Systematic review and meta-analysis (protocol)

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1. Introduction

Metformin, a commonly used for antidiabetic drug, has shown an anticancer effect in patients with breast cancer.^{1,2} Several in vitro studies have been reported on the anticancer pathways. The first, metformin activates AMP-activated protein kinase (AMPK) and subsequently leads to inhibition of mammalian target of rapamycin (mTOR) signaling to cell growth.^{3,4} Second, a systematic review reported that metformin inhibited signal transducer and activator of transcription (STAT3) activation and induced apoptosis.⁵

Findings on the effect of metformin are inconsistent in breast cancer clinical researches. Several studies showed that the combination therapy of metformin and chemotherapy had a major antitumor effect.^{6,7} In patients with HER2 (+) hormone receptor-positive breast cancer from ALTTO trial cohort, the patients with diabetes treated with metformin had better disease free survival, distant disease-free survival, and overall survival (OS) than the patients with diabetes who were not treated with metformin.⁸ On the other hand, several randomized controlled studies revealed that metformin

and concomitant use of chemotherapy or endocrine therapy showed no significant efficacy on survival.⁹⁻¹¹ There is no systematic review of adding metformin to chemotherapy or endocrine therapy for breast cancer patients. The purpose of this systematic review is to elucidate the efficacy of metformin and concomitant use of chemotherapy or endocrine therapy in patients with breast cancer.

2. Research question

P: patients with breast cancer

I: metformin and concomitant use of chemotherapy or endocrine therapy

C: chemotherapy or endocrine therapy with or without placebo

O: OS, progression free survival (PFS), rate of hypoglycemia, adverse events (AEs), relapse-free survival (RFS)

3. Method

3.1 Inclusion criteria of the articles for the review

3.1.1 Type of studies

We will include individual randomized controlled trials. We will not apply language or country restrictions. We will include all papers including published, unpublished articles, abstract of conference, and letter. We will exclude crossover trials, quasi-experimental studies, and quasi-randomized trials. We will not exclude studies based on the observation period.

3.1.2 Study participants

Inclusion criteria:

patients with breast cancer receiving chemotherapy or endocrine therapy, Eastern Cooperative Oncology Group (ECOG) performance status (PS) 0-2, over 18 years old women, and life expectancy at least 3 months

Exclusion criteria:

inability to take metformin or placebo, a history of allergy or adverse reactions to metformin

3.1.3 Intervention

metformin and concomitant use of chemotherapy or endocrine therapy

3.1.4 Control

chemotherapy or endocrine therapy with or without placebo

3.2 Type of outcomes

3.2.1 Primary outcomes

1. OS

Definition: from the day of randomization to the date of death from any cause or loss to follow-up

Period: during follow up period

2. progression free survival (PFS)

Definition: from the day of randomization to the date of disease progression, death from any cause, or loss to follow-up

Period: during follow up period

3. rate of hypoglycemia

Definition: blood glucose level was defined as ≤ 70 mg/dL

Period: during the follow up period

3.2.2 Secondary outcomes

1. Severe Adverse Events (SAEs)

Definition: $\geq$ grade 3 toxicity in common terminology criteria for adverse events (CTCAE) version 5.0 or previous versions.

Period: during follow up period

2. relapse-free survival (RFS)

Definition: from the day of randomization to the date of breast cancer recurrence, which includes local, regional, or distant recurrence or death by any cause. Patients receiving adjuvant chemotherapy or endocrine therapy were eligible.

Period: during follow up period

3.3 Search method

3.3.1 Electronic search

We will search the following databases:

1. the Cochrane Central Register of Controlled Trials (CENTRAL);
2. MEDLINE via Ovid;
3. EMBASE;

See Appendix 1, 2, and 3 for the search strategies.

3.3.2 Other resources

We will also search the following databases for ongoing or unpublished trials:

1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP)

See Appendix 4 for the search strategies.

2. ClinicalTrials.gov

See Appendix 5 for the search strategies.

We will check the reference lists of studies, including international guidelines^{12,13} as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.4 Data collection and analysis

3.4.1 Selection of the studies

Two independent reviewers (K. Morio and Y. Kurata) will screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. We will make contact with the original authors if relevant data is missing. Disagreements between the two reviewers will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (Y. Kataoka).

3.4.2 Data extraction and management

Two reviewers (K. Morio and Y. Kurata) will perform independent data extraction of the included studies using standardized data collection form. The form will include the information on study design, study population, interventions, and outcomes. Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (Y. Kataoka). For the data extraction, we will use a pre-checked form using 10 randomly selected studies.

3.5 Assessment of risk of bias in included studies

Two reviewers (K. Morio and Y. Kurata) will evaluate the risk of bias independently using the Risk of Bias 2.¹⁴ Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer (Y. Kataoka) will be acting as an arbiter, if necessary.

3.6 Measures of treatment effects

We will pool the mean difference (MD) and the 95% confidence intervals (CIs) for the following the continuous variable: rate of hypoglycemia

We will pool the hazard risk and the 95% confidence intervals (CIs) for the following survival time: OS, PFS, RFS

Neoadjuvant, adjuvant and metastatic/recurrent chemotherapy or endocrine therapy will be evaluated.

We will summarize adverse events based on the definition by the original article, but we will not perform meta-analysis.

3.7 Unit of analysis issues

None

3.8 Handling of missing data

3.8.1 Missing participants

For dichotomous data

We will perform the intention-to-treat (ITT) analysis for all dichotomous data. We will also include missing participants for analysis.

For those dropped out from the study early, they are assumed to have the same rates of negative outcome on the basis of the rates of those who completed the study.

We will underestimate the treatment effect by this method. We will describe how to impute missing data in each study. We will conduct the sensitivity analysis for imputation for missing data.

For continuous data

For continuous data, we will not impute missing data based on the recommendation by Cochrane handbook.¹⁵ We will perform meta-analysis about the available data in the original study.

3.8.2 Missing values

We will ask missing values to the original authors.

3.8.3 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman.¹⁶ If we don't know these values when we contact the authors, standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook¹⁵, or validated method.¹⁶ Validity of these methods will be analyzed by sensitivity analysis.

3.9 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I² statistic (I² values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). When there is substantial heterogeneity (I² > 50%), we will assess the reason of the heterogeneity. Cochrane Chi²test (Q-test) will be performed for I² statistic, and P value less than 0.10 will be defined as statistically significant.

3.10 Assessment of reporting bias

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. We will assess the potential publication bias by visual inspection of the funnel plot. Egger test will be performed as well. We will not conduct the test when we find less than 10 trials or trials which have similar sample size. We will assess the potential publication bias by visual inspection of the funnel plot.

3.11 Meta-analysis

Meta-analysis will be performed using Review Manager software (RevMan 5.4). We will use a random-effects model.

3.12 Subgroup analysis

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. (For participants) According to hormone receptor and human epidermal growth factor receptor 2 (HER2) status, patients were divided into three groups:

①neoadjuvant chemotherapy or endocrine therapy: ER/PgR (+) HER2 (-), ER/PgR (+/-) HER2 (+), triple negative breast cancer (TNBC)

②adjuvant chemotherapy or endocrine therapy: ER/PgR (+) HER2 (-), ER/PgR (+/-) HER2 (+), TNBC

③metastatic/recurrent chemotherapy or endocrine therapy: ER/PgR (+) HER2 (-), ER/PgR (+/-) HER2 (+), TNBC

2. (For participants) type 2 diabetes mellitus vs non-diabetes mellitus

3. (For participants) body mass index (BMI) ≥ 25 kg/m² vs BMI < 25 kg/m²

4. (For intervention) take metformin ≥ 1.5 years vs < 1.5 years

3.13 Sensitivity analysis

We will undertake sensitivity analyses for the primary outcome to assess whether the results of the review are robust to the decisions made during the review process. We plan to examine the effects of the review findings of:

1. Exclusion of studies using imputed statistics.

2. Missing participants: verify the robustness of the results by seeking informative missingness odds ratios.

3. Only the participants who complete the study with complete data.

4. Summary of findings table

Summary of findings table will be made for the following outcome based on the Cochrane handbook.¹⁶ We will include grading to evaluate the quality of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach for each Summary of findings table.¹⁷

1. OS

2. PFS

3. rate of hypoglycemia

4. AEs

5. RFS

5. Conflict of Interest

The authors declare no conflicts of interests.

Appendix 1: CENTRAL search strategy

1. MeSH descriptor:[breast neoplasms]explode all trees
2. "breast cancer":ti,ab,kw
3. 1 OR 2
4. MeSH descriptor:[metformin]explode all trees
5. metformin:ti,ab,kw
6. 4 OR 5
7. 3 AND 6

Appendix 2: MEDLINE(via PubMed) search strategy

- #1 "breast neoplasms"[mh]
- #2 "breast cancer"[tiab]
- #3 #1 OR #2
- #4 metformin[mh]
- #5 metformin[tiab]
- #6 #4 OR #5
- #7 #3 AND #6
- #8 "randomized controlled trial"[pt]
- #9 "controlled clinical trial"[pt]
- #10 randomized[tiab]
- #11 placebo[tiab]
- #12 "drug therapy"[sh]
- #13 randomly[tiab]
- #14 trial[tiab]
- #15 groups[tiab]
- #16 #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15
- #17 animals[mh] NOT humans[mh]
- #18 #16 NOT #17
- #19 #7 AND #18

Appendix 3: EMBASE (via EMBASE.com) search strategy

1. "breast cancer":ab,ti. OR "breast cancer"/exp
2. metformin:ab,ti. OR metformin/exp
3. ('sensitivity'/exp OR sensitivity) AND maximizing AND ('strategy'/exp OR strategy) AND random*:ab,ti OR ((clinical NEXT/1 trial*):de,ab,ti) OR 'health care quality'/exp OR 'health care quality')
4. 1 AND 2 AND 3

Appendix 4: ICTRP search strategy

1. Conditions: (breast cancer OR breast neoplasms)
2. Intervention: metformin
3. 1 AND 2

Appendix 5: ClinicalTrials.gov search strategy

1. Condition or disease: (breast cancer OR breast neoplasms)
2. Intervention: metformin
3. 1 AND 2

Appendix 6

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